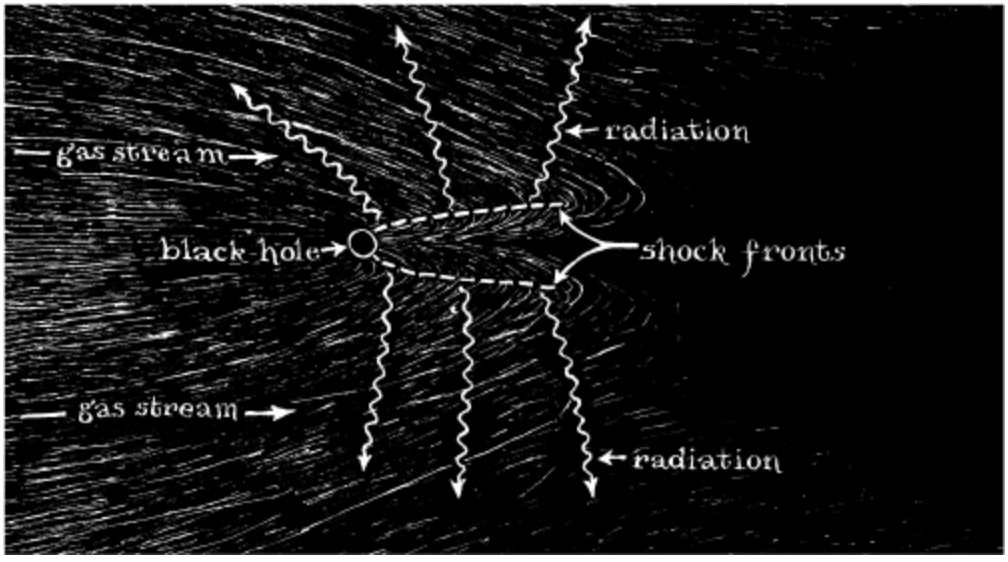
up with a new list of eight black-hole candidates. Unfortunately, in all eight cases, Trimble could invent a semi-reasonable non–black-hole explanation for why the companion was so dark. Today, a quarter century later, none of our candidates has survived. It now seems likely that none of them is truly a black hole.

Zel'dovich knew, when he conceived it, that this binary star method of search was a gamble, by no means assured of success. Fortunately, his brainstorming on how to search for black holes produced a second idea—an idea conceived simultaneously and independently, in 1964, by Edwin Salpeter, an astrophysicist at Cornell University in Ithaca, New York.

Suppose that a black hole is traveling through a cloud of gas—or, equivalently, as seen by the hole a gas cloud is traveling past it (Figure 8.4). Then streams of gas, accelerated to near the speed of light by the hole's gravity, will fly around opposite sides of the hole and come crashing together at the hole's rear. The crash, in the form of a *shock front* (a sudden, large increase in density), will convert the gas's huge energy of infall into heat, causing it to radiate strongly. In effect, then, the black hole will serve as a machine for converting some of the mass of infalling gas into heat and then radiation. This "machine" could be highly efficient, Zel'dovich and Salpeter deduced—far more efficient, for example, than the burning of nuclear fuel.



8.4 The Salpeter–Zel'dovich proposal for how to detect a black hole.